Naor Larian

FPGA & RTL | ELECTRICAL ENGINEERING STUDENT

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+ SUMMARY

Final-year Electrical & Electronics Engineering student (expected 2027) with hands-on VHDL and Vivado experience. Designed and implemented a working FPGA system on Artix-7 hardware, integrating ESP32 control over UART with real-time VGA graphics. Nearly four years of technical experience in live satellite communication systems.

+ FPGA PROJECTS

● 2026 / Cyber Gladiators, Independent FPGA Project

- Designed a modular **VHDL game engine** with 640 × 480 VGA output, sprite rendering, FSM-based control, collision detection, and gameplay logic.
- Integrated an **ESP32 controller through UART** and demonstrated the complete design running on a physical Digilent Basys 3 (Artix-7) board.
- **Verified RTL modules through simulation**, synthesized and implemented the design in Xilinx Vivado, and debugged UART-to-VGA integration on physical FPGA hardware.

● 2026 / RGB Median Filter, Afeka Team Project

- Developed **internal VHDL components** and integrated FSM control, data buffers, and RAM/ROM handling for a 3 × 3 image median filter.
- Verified module and top-level functionality using RTL simulations and image test data.

+ EXPERIENCE

● 2022 - Present / AMOS-Spacecom, Technical Support Engineer

- Investigate and troubleshoot **15–25 technical incidents per week** across live satellite systems, correlating RF spectrum measurements, modem indicators, and link performance data.
- Support approximately **10–15 CPI tests per month**, performing operator-side RF measurements and guiding customers through testing procedures.
- Developed and refined **technical SOPs** to improve troubleshooting consistency and make procedures and critical information easier to find.

+ TECHNICAL SKILLS

FPGA & RTL: VHDL, FSM, RTL simulation, testbenches, Vivado, Artix-7 / Basys 3

Interfaces & systems: UART, ESP32, VGA, RAM/ROM, hardware integration

Programming & tools: C, Python, MATLAB/Simulink (academic); spectrum analyzer (professional)

+ EDUCATION

2027 (expected) / B.Sc. Electrical & Electronics Engineering

Afeka Tel Aviv Academic College of Engineering. Computers & Communications specializations.

2022 / Practical Engineer Diploma, Electrical Engineering

Handessaim Tel Aviv, graduated with honors.

+ RELEVANT STUDIES

Advanced Hardware Design, Digital Signal Processing, Communication Systems, Signals & Systems.

+ LANGUAGES

Hebrew / Native

English / Professional working proficiency